

Research Assignment 2 - Due 26 Sept 2020

Section 01: Foundational concepts of Analytics

1) **Business Analytics**: The concept refers to statistical methods and computing technologies for processing, mining and visualizing data to uncover trends, relationships and insights that enable better of business - decision making (Evans, 2020)

Real Life Example → An e-commerce furniture shop uses business analytics to draw insights on how the shop performs between christmas holidays and easter holidays.

2) **Data Analytics**: It is the process of collecting, transforming and organizing data in order to draw conclusions, make predictions and drive informed decision making (Davenport and Harris, 2017)

example → A clinical research organisation uses data analytics to track subjects enrollment across various multi-sites.

3) **Business Intelligence (BI)**: It is a set of technological processes that focus on collecting, querying, and analyzing historical and current organization data to describe what is currently happening within the organization (Sharda et al., 2020)

Example → A logistics company uses BI to track deliveries across regional warehouses.

04) Data-Driven Decision Making: It is a management approach whereby decisions are guided by data-analysis rather than intuition. The decisions are based more on verified operational metrics (McAfee and Brynjolfsson, 2012)

example → A Research Organization deciding on which studies to do more off based on the successful completion of studies with no deviations.

05) Big Data: It refers to extremely large, complex data^{sets} that traditional data processing software tools cannot easily manage, store, or analyze (Monyica et al) 2011

example → Real time GPS traffic navigation.

06) Descriptive Analytics: It is the process of using current and historical data to summarize, organize and identify trends within a specific datasets. It answers the basic question: "What happened?" (Evans; 2020)

example → Financial statements analysis of a business, to give them a holistic view of what the company's current expenditure.

07) Diagnostic Analytics: It analysis historical datasets to uncover or examine root causes, patterns and reasons behind past events or performances. It answer the question: "Why did it happen?"

example → A hospital analysis why specific patients group have long recovery time. The historical data then shows that there is lack of Post-discharge follow up.

08) Predictive Analytics - It is a branch of advanced analytics that uses historical data, statistical modelling, and machine learning to predict future events and trends. It forecasts what might happen to better guide the choices (Davenport and Harris, 2017)

example → In epidemiology it is used during a disease outbreak to identify which demographic will be at risk for disease progression based on historical data.

09) Prescriptive Analytics : It is used to analyze data to identify patterns which can be used to make predictions and recommend the best course of action (Evans, 2020)

example → In Healthcare, recommending specific patient treatments based on medical history, current symptoms and recovery chances.

10) Data Literacy : It is the ability to read, understand, work with, analyze and communicate effectively with data. It is the ability to ask the right questions and spot data discrepancies within a dataset (Harvard Business Review, 2017)

example → You look at the weekly/monthly breakdown of transactions in your banking app.

Section 02: Data, Measurement & Reporting

- 11) **Structured Data**: It is ~~an~~ data highly organized information in the form of defined rows and columns. Because it is standardized, it can be queried, and analyzed easily. (Sharda et al; ²⁰²⁰~~2002~~)

example → A bank database holding customer account numbers, transaction amounts and dates in a standardized table.

- 12) **Unstructured Data**: It is data that lacks a defined model or schema. It is often harder to collect and analyze using conventional databases (Munika et al; 2011).

example → A brand analyzing feedback on social media posts.

- 13) **Data Warehouse**: It is a centralized repository that consolidates data from various, multiple operational systems across an enterprise. This data is used for analytical reporting. The data warehouse cleans and organizes historical data specifically for executive querying. It uses automated process such as ETL (Extract, Transform & Load) (Sharda et al; 2020)

example →
↳ A large retail business/chain collects information from online website orders.

14) Data Mining : It refers to the computational process of analyzing large dataset to uncover hidden patterns, data anomalies, correlations and trends. It turns raw data into practical knowledge using tools such as statistics and machine learning (Han et al; 2011)

example → A grocery store uses loyalty cards to track customer purchases.

15) Data Visualization : It is the graphical representation of data using visual elements such as charts, graphs, maps. Data visualization tools makes it easier to see and understand trends, outliers and patterns in data (few, 2012)

example → Fitness watches use circle rings to show step count, distance covered and calories lost.

16) Dashboard : It is a real-time visual tool that tracks, analyzes and displays key metrics and data as they update live (few, 2012)

example → Finance dashboards monitoring daily business cash flow, profit margins and sales.

17) Metric : It is a system of measurement, quantifiable measure used to track, compare and evaluate performance of a specific business process or activity (Parmenter, 2015)

example → The total number of raw page views recorded on company blog during a week.

18) Key Performance Indicator (KPI): It is a specific metric that directly measures progress towards a primary strategic goal. KPIs are often narrow and focused on vital business objectives (Porter 2015)

example → Business cash runway: Calculate how many months your business can survive before running out of cash.

19) Benchmark: It is a process within a company comparing its performance, ~~against~~ products, or internal processes against industry leaders or best practices to find areas of improvement (Investopedia, 2023)

example → A hotel chain compares its 75% occupancy rate against the regional industry benchmark of 60%.

20) Segmentation: It is the process of dividing a broad customer base or dataset into different sub-groups based on shared demographics, behaviors or purchasing patterns. This allows organizations to target strategies differently (Kotler and Kotler, 2016)

example → Online fashion retailer dividing its customer base list into frequent buyers and seasonal buyers.

Section 03: Customer & Sales KPIs

21) **Conversion Rate**: It is the percentage of people who complete a specific goal out of a total group. In business, it ~~is the~~ ^{measures} how effectively a website or an ad turns visitors into customers (Sales Hubspot, 2023)

example $\rightarrow$ ~~A business running ads on meta in a~~
An e-commerce business store tracks that out of 10000 store visits, 300 customers completed a purchase.
~~month to attract new customers.~~

22) **Customer Acquisition Cost (CAC)**: It refers to the total expenditure required to acquire a single new customer over a specific period. This includes all marketing spend, sales expenses and software tools (Harvard Business Review, 2007)

example $\rightarrow$ A business running R5000 ads on meta in a month to attract ⁵⁰⁰ new customers which results in R100 per customer

23) **Customer Lifetime Value (CLV / LTV)**

✓ It is defined as the total worth of or profit from a customer to a business over the entirety of their relationship. This looks at how valuable a customer is to an organization as a whole and how much capital to allot toward acquisition and retention (Forris et al, 2010)

example $\rightarrow$ Monthly gym recurring subscription of R200 over a 24 month retention period

$$\textcircled{*} \text{ CLV} = (\text{Average Order Value} \times \text{Purchase frequency} \times \text{Customer Lifespan}) \times \text{Gross Margin}$$

$\textcircled{*}$ **Customer Lifetime Value (CLV)**: Customer Acquisition Cost (CAC)

example $\rightarrow$ local subscription box service.

CAC: R500 spent on digital ~~ads~~ ^{ads} per customer

CLV: R1500 total profit over the ~~subscriber~~ ^{subscriber} lifespan

CLV : CAC

1500 : 500

= 3 : 1 ratio.

3:1 ratio is the benchmark for a profitable business whereas if the ratio is ^{below} 1:1, the business would be losing money for every customer acquired. Then 1:1, 2:1 ratio means the business is barely breaking even after operating costs.

25) Customer Retention Rate $\%$: Refers to the percentage of existing customers a business keeps over a specific time period (Forris et al, 2020)

$$\text{Customer retention rate} = \left(\frac{E - N}{S} \right) \times 100$$

E = number of customers at the end of the period

N = number of new customers acquired during the period

S = Number of customers at the start of the period.

→ Why is Customer ~~Ratio~~ ^{Retention} rate important: Lower costs keeping existing clients after costs significantly less than acquiring new ones.

24) Churn rate: It measures the percentage of customers who stop using a company's product or service within a specific time period. It shows how well a company keeps its customers. (Forris et al, 2020)

Example → A business productivity app begins the quarter with 400 paying corporate accounts. Over the period of 6 months, 40 companies downgrade to a lower tier or cancel their accounts entirely.

26) Net Promoter Score (NPS): It is a customer experience metric that measures the probability / likelihood that your customers may recommend your business or service to friends or colleagues (Fred Reichheld, 2003)

example → ~~A business having a survey after a customer completes an online shopping~~
A business asking a customer to complete a survey after the successful delivery of a service.

27) Average Order Value (AOV): It refers to tracking the average amount of money a customer spends each time they place an order on a website or mobile app (Investopedia, 2023)

example → Revenue of ₹10000 from 200 orders yields an AOV of ₹50

28) Sales Funnel: It is a step-by-step model entailing a customer's journey from first discovering the brand to purchasing a product. (Kotler and Keller, 2016)

Section 04 Marketing and Campaign Analytics.

29) Marketing Campaign : It is a set of strategic, planned activities that are aimed ^{at} ~~to~~ promoting a company's product or service or brand objective across media channels over a set period (Kotler and Keller, 2016)

example → A healthcare company promoting a medical device ~~across~~ across different social media channels.

30) Post-campaign Analysis : It is an evaluation conducted after ~~the~~ running a marketing campaign. It involves a formal review of marketing, advertising or public relations efforts after it ends to measure results, determine return on interest (ROI) and further guide future strategy (Farris et al, 2010)

example → A healthcare company providing a detailed real-time feedback analysis on the medical device marketing campaign.

31) Incremental Revenue → It is when a business earns an extra revenue directly generated by a campaign over and above what would have been earned without the marketing campaign (Farris et al, 2020)

Example $\Rightarrow$ A retail company normally earning R250 000 a week generates R20370 000 during a social media campaign thus making

32) Return on Investment (ROI): It is a financial ratio that measures the net profit of an investment relative to its total costs (Investopedia, 2023)

Example $\Rightarrow$ A company spending R10 000 on a product marketing that then generates R15 000 in net profits.

33) Return on Ad Spend (ROAS): It measures the gross revenue of business generated for every unit spent on advertising. (Bailey, 2025)

Example $\Rightarrow$ Spending R1000 on Google search Ads to produce R5000 ROAS. in sales yields a ROAS of 5.0

$$\begin{aligned}\text{ROAS} &= \text{Revenue Generated} / \text{Ad spend} \\ &= 5000 / 1000 = 5.0\end{aligned}$$

34) Click-Through Rate: It is the percentage of people who click on a link or advertisement after seeing it. (Investopedia, 2023)

Example $\Rightarrow$ An online store receives 200 clicks from 1000 page views.

35) Bounce Rate: Refers to the percentage of website visitors that leave the site without interacting with any other pages or taking further action (Google Analytics help, 2023)

example → A landing page of a business where 70 of 100 ~~visitors~~ visitors ^{exit} the website without clicking on links.

36) A/B Testing: It is a randomized experiment comparing two versions A (Control) and B (Variation) of a digital asset to see which one performs better. This experiment is performed ~~later~~ simultaneously (Kohavi et al, 2020)

example → An app team compares a green checkout button against a red checkout button to determine which one yields a higher click conversion rate.

37) Lead Generation: It is a marketing process of attracting interest from prospective customers and capturing their contacts to move them into a sales pipeline (HubSpot, 2023)

example → Offering a free downloadable ebook or research report in return of an email address.

38) Marketing Attribution: It is the process of determining which marketing touchpoints drove a buyer's journey and the financial credit for bringing in sales (Forris et al, 2020)

example → Using a multi-touch attribution to allocate sales credit proportionally between Facebook ad, and email newsletter